

Iyaad Amsyar Syahrizal

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EDUCATION

Swansea University | Wales, UK

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Bsc. in Software Engineering, 2nd Year / 2:1

- **Courses:** Robotics, Algorithms & Analysis, Modern Web Development, CyberSecurity, Mobile Operating Systems, Data Structures,, Foundations of Programming, Files and Databases

TECHNICAL SKILLS

Languages (ordered by proficiency): Java, Python (Pandas, Matplotlib, Plotly, pytest), JavaScript, C#, Haskell, Prolog, TypeScript

Tools and Frameworks: Node.js, React.js, Git

PROJECTS

Jewel Thief — Group Game Project (Java + JavaFX)

Developed a tile-based puzzle game as part of an 8-person team in the CS-230 Software Engineering module

- Built the data persistence system using Java and Jackson (JSON), handling player profiles, save/load game state, and high score tracking
- Designed a clean architecture with CRC cards and UML diagrams, applying the Singleton pattern for the persistence manager
- Implemented JUnit 5 tests for JSON round-trips, save integrity, and edge cases (corrupted files, version migration)
- Practised Agile workflows including iterative development, integration with teammates' game logic and UI code

Chess Game with AI

- Built a fully playable chess game in C# featuring an AI opponent
- Implemented game logic, move validation, and basic AI decision-making

Built a compiler in Java and C

Ransomware Threat Model — ANOther Ltd

- Modelled a ransomware attack scenario using **attack trees** (AND/OR semantics) in AD Tool
- Mapped adversary tactics to the **MITRE ATT&CK framework**, identifying techniques such as phishing (T1566), data encryption for impact (T1486), and double extortion strategies
- Scored attack vectors using the Attack Potential methodology (time, specialist knowledge, knowledge of target, window of opportunity, equipment)

RELEVANT EXPERIENCE

Discord Community Moderator

Managed multiple servers with 2,000+ members, maintaining engagement and enforcing community standards

Assistant Teacher — Coding & Chess Clubs

Mentored peers in programming fundamentals and chess strategy.

Organised and ran school-wide chess tournaments

Badminton Coach

Coached players who went on to compete in inter-school tournaments

